DATA FEASIBILITY AUDIT · TRANSKNIGHTS PRODUCTION DB

FreightStream Capability Map

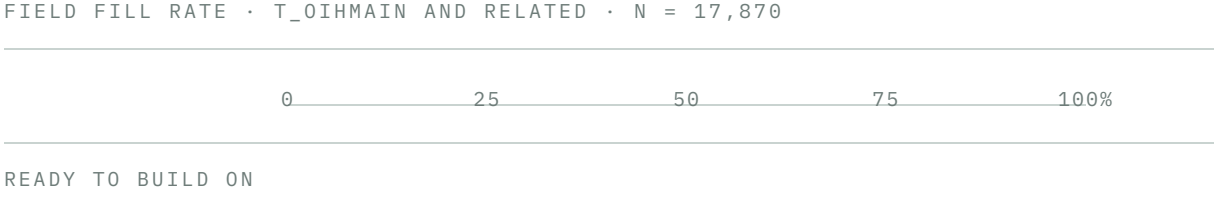
We tested all five goals — operational oversight, staff assessment, cost reduction, early problem detection and incident prevention — against the real data. One sentence covers the result: **the operational milestone data is almost entirely empty, but the record of who touched what and when is complete.** So every capability has to be built on the audit trail and the financial ledger, not on operational milestones. That one fact decides whether each function below succeeds or fails.

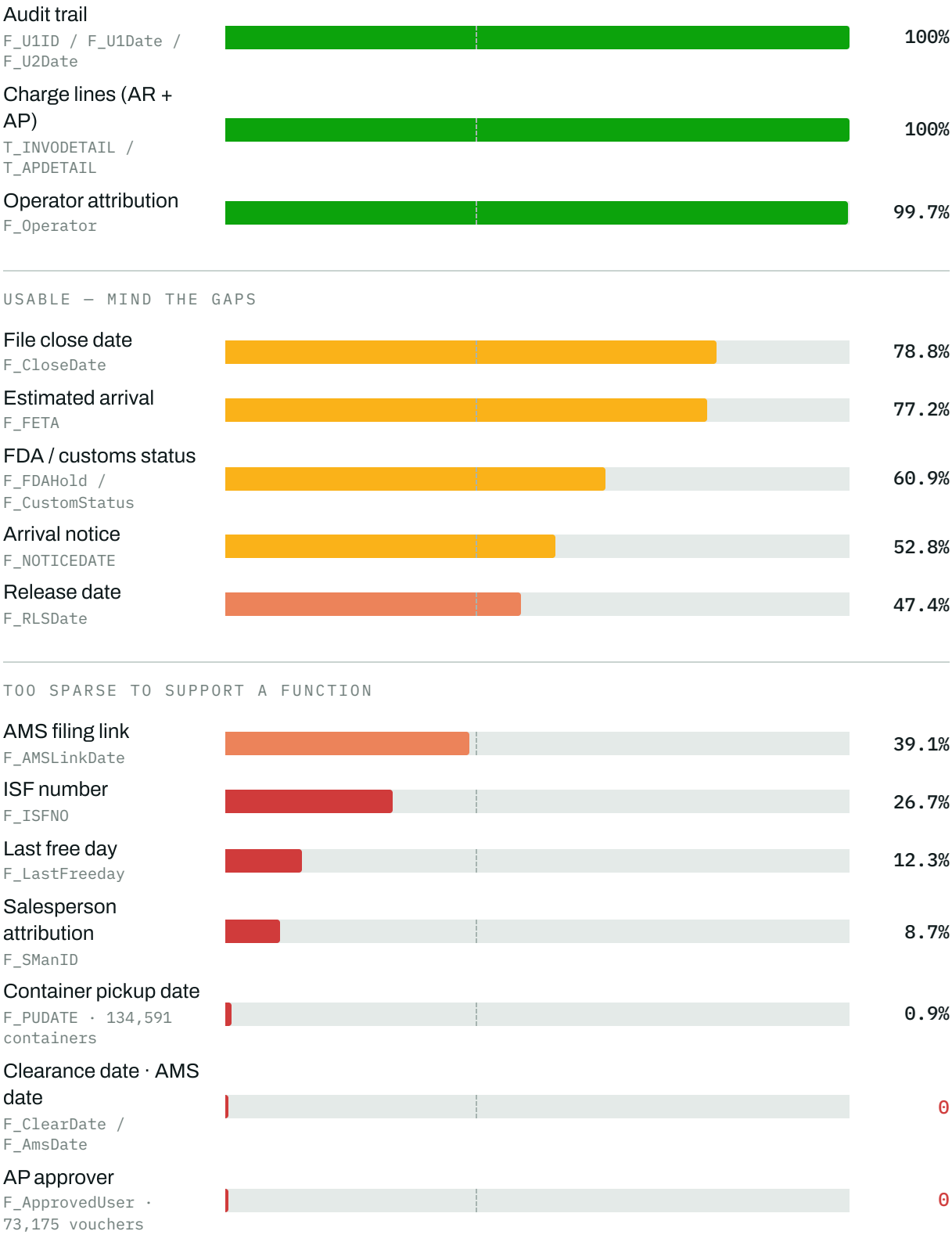
SOURCE	SIZE	SPAN	AUDIT SAMPLE
MSSQL / TRANSKNIGHTS	233 tables · 12.8 GB	2007 – 2026	17,870 files · since 2024
ACCESS			
Read-only · verified 09-04			

01 — AUDIT FINDINGS

A field exists. That does not mean it holds data.

Within one table some fields are 100% populated and others hold nothing at all. Below is the measured fill rate across 17,870 ocean import files opened since 2024 — it decides which functions can ship today and which need a process change first.




Audit trail
F_U1ID / F_U1Date / F_U2Date
100%
Charge lines (AR + AP)
T_INVODETAIL / T_APDETAIL
100%
Operator attribution
F_Operator
99.7%
USABLE — MIND THE GAPS
File close date
F_CloseDate
78.8%
Estimated arrival
F_FETA
77.2%
FDA / customs status
F_FDAHold / F_CustomStatus
60.9%
Arrival notice
F_NOTICEDATE
52.8%
Release date
F_RLSDate
47.4%
TOO SPARSE TO SUPPORT A FUNCTION
AMS filing link
F_AMSLinkDate
39.1%
ISF number
F_ISFNO
26.7%
Last free day
F_LastFreeday
12.3%
Salesperson attribution
F_SManID
8.7%
Container pickup date
F_PUDATE · 134,591 containers
0.9%
Clearance date · AMS date
F_ClearDate / F_AmsDate
0
AP approver
F_ApprovedUser · 73,175 vouchers
0
Ready (>95%)
Usable, gaps to handle (50–80%)
Marginal (25–50%)
Unusable (<25%)
Dashed line = 40% usability floor

The worst finding is not low coverage — it is a feature never switched on

Across 73,175 AP vouchers, `F_Approved` is `'0'` on every one and the approver field is empty throughout — meaning **no payment of any size carries a system-level approval record**. This is an internal-control gap, not an analytics one. It deserves its own agenda item.

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02 — COST & EFFICIENCY

Three pots of money already located

None of these need new data captured. All can be computed — and managed — from what the database already holds.

BILLING LAG · 2025

35.1 days

Average time from opening a house B/L to issuing the invoice. 51% of 7,425 invoices took more than 30 days.

WORKING CAPITAL UNLOCKED

≈$3.43 M

Cutting billing to 15 days takes 20 days off DSO. Estimated against 2025 billings of $62.65M.

DUPLICATE PAYMENT EXPOSURE

252,904 USD

Same vendor, same invoice number, entered more than once: 283 groups, 808 surplus records (since 2024).

DEMURRAGE NOT PASSED ON

945,298 USD

107 files were invoiced to the customer, with no demurrage line on the invoice (2024).

CASH CONVERSION CYCLE · FILE OPENED → CASH RECEIVED

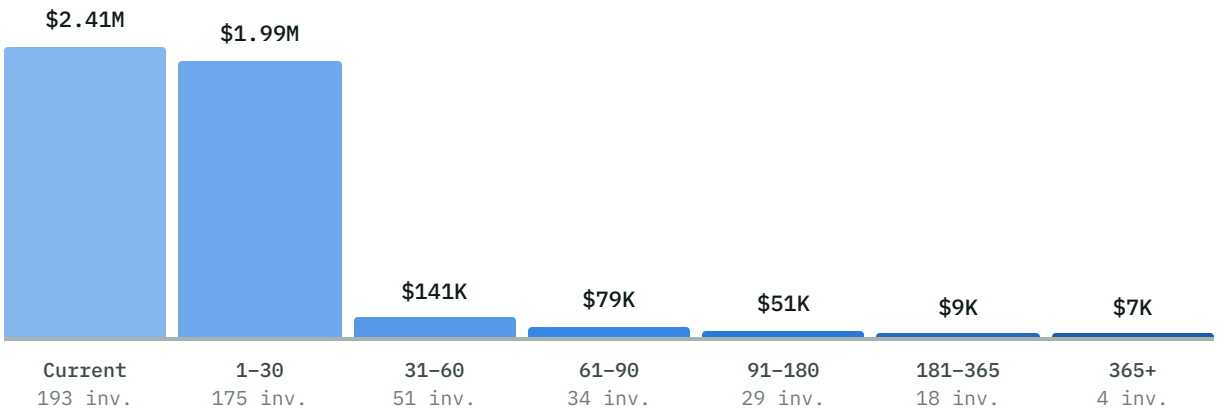


File → invoice
Self-inflicted, fully controllable

Invoice → cash
Customer-held, needs negotiation

The full cycle runs about **53 days**, and **two-thirds of it is internal process**. Collections need negotiation; billing needs only a process change. That is why billing timeliness comes first — more leverage than collections, and it asks nothing of the customer.

OPEN RECEIVABLES BY AGE · 504 INVOICES · $4,691,631 · AS OF 2026-09-04



The ledger itself is healthy — debt over 90 days is just **1.4%**. But note: 57% of all invoices carry nominal terms of zero days and the on-time rate is only 20%, so **judging overdue against the nominal due date turns 80% of invoices red**. The workbench has to measure overdue against each customer's own payment baseline instead.

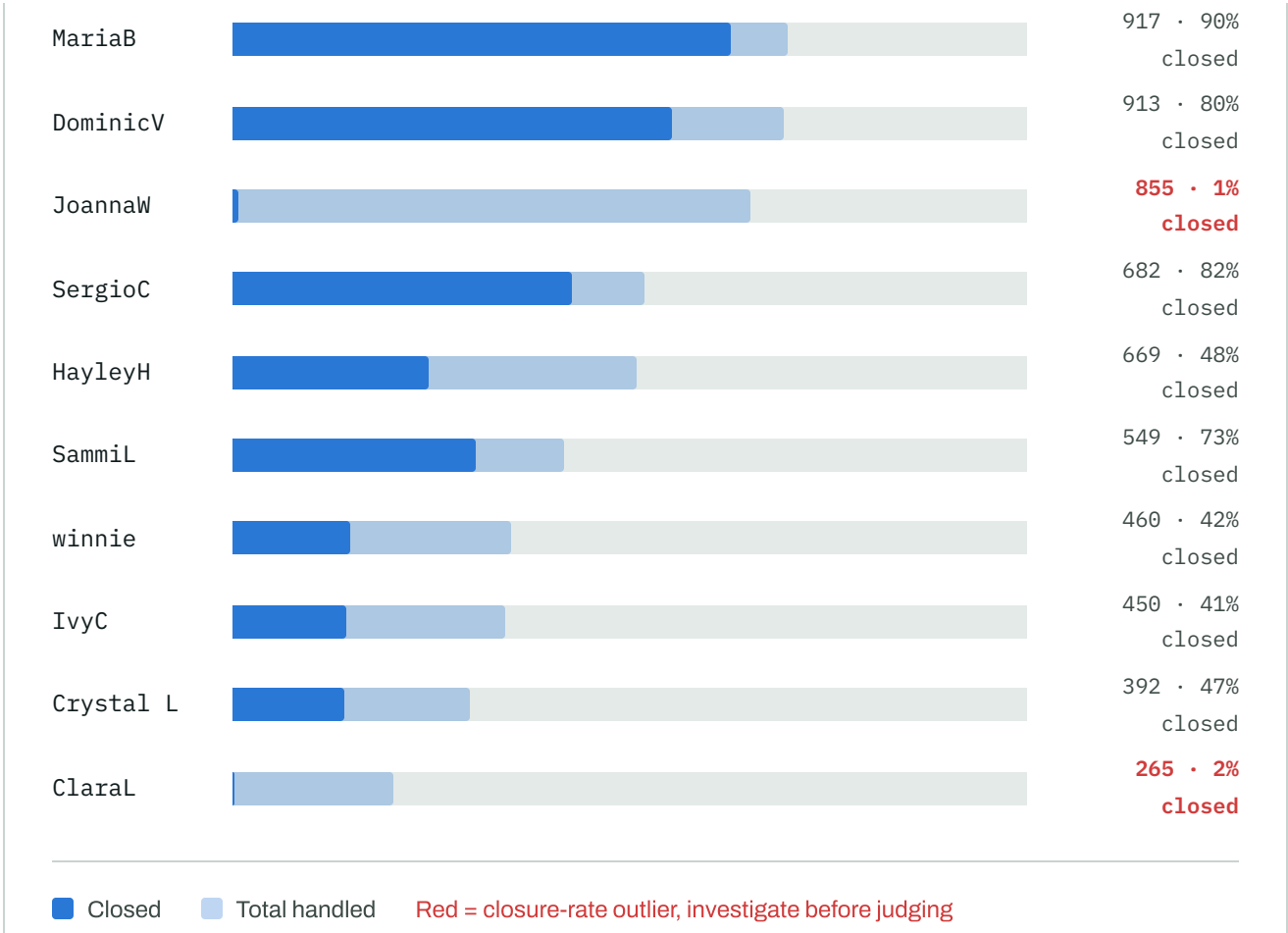
03 — STAFF ASSESSMENT

The foundation is there — and one metric trap

`F_Operator` covers 99.7% of files, so workload, timeliness and quality are all computable. Below: files handled per operator in 2025.

FILES HANDLED PER OPERATOR · 2025 · SOLID = CLOSED





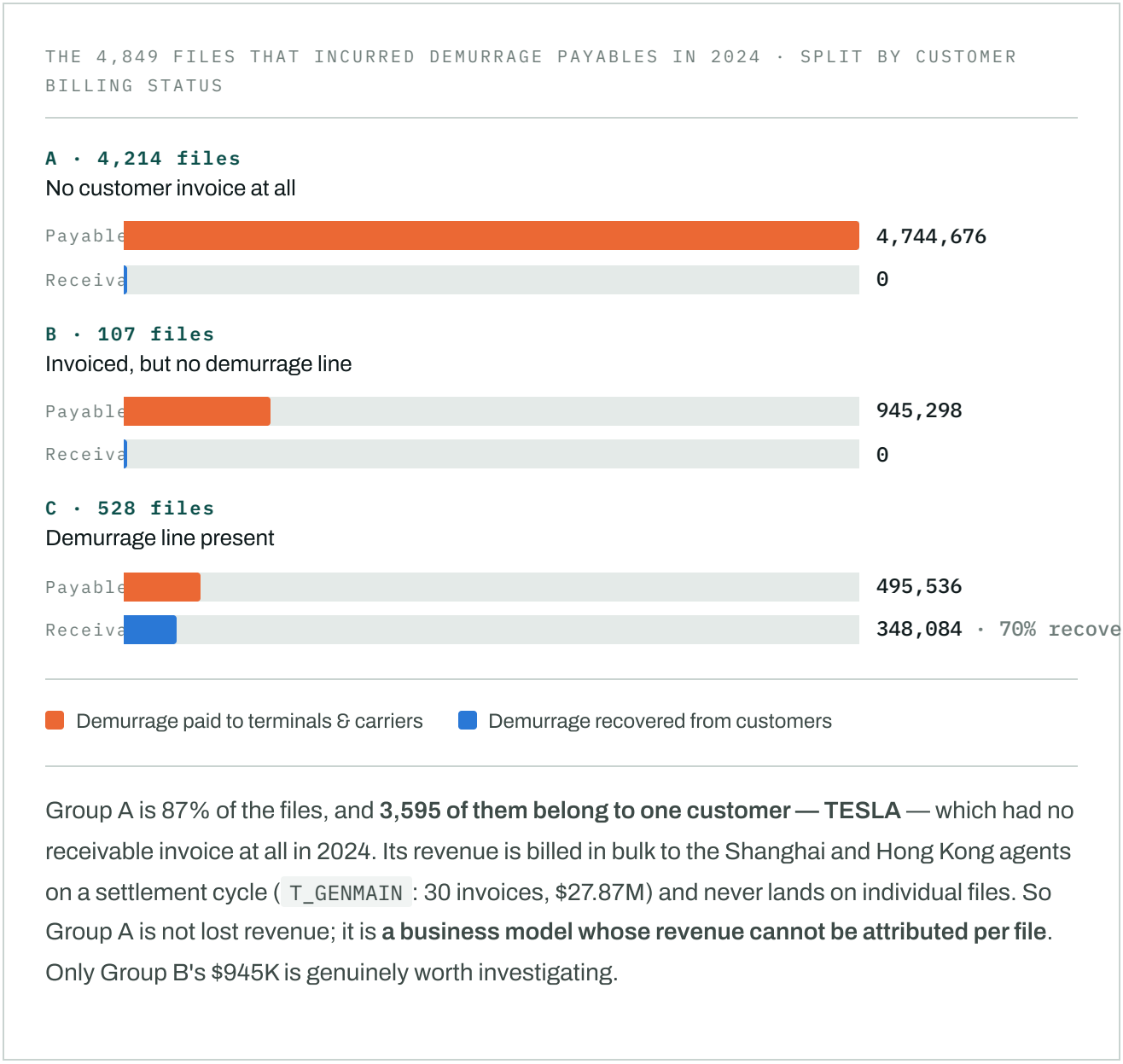
Do not use days-to-close as a KPI

Files close after **340 days** on average and 89% take over 60 days — closing is a finance batch action, not a measure of operational speed. As a KPI it manufactures false problems and pushes staff to game a number that means nothing. **Use billing lag and cost-booking lag instead**; those are what an operator actually controls. Same caution for JoannaW and ClaraL: ask whether their role differs before treating it as performance.

04 — DEFINITION TRAPS

Why per-file metrics go wrong

This section was paid for with a real mistake: a finding that looked like an $11.2M leak turned out not to exist.



Group A is 87% of the files, and **3,595 of them belong to one customer — TESLA** — which had no receivable invoice at all in 2024. Its revenue is billed in bulk to the Shanghai and Hong Kong agents on a settlement cycle (`T_GENMAIN` : 30 invoices, $27.87M) and never lands on individual files. So Group A is not lost revenue; it is **a business model whose revenue cannot be attributed per file**. Only Group B's $945K is genuinely worth investigating.

Segment first, then analyse

Of receivables since 2024, direct customers billed per file account for **57%** and agency business billed per settlement cycle for **43%**. Per-file profitability, lane margin, customer margin, demurrage recovery — none of these hold for the agency segment. The two need two separate definitions; **mixed into one report they will be wrong**. Enforce it as a dimension in the data layer rather than relying on an analyst to remember.

05 — CAPABILITY LIST

Seven capabilities, ranked by speed-to-value over dependency

The first six need no new data capture at all. The seventh requires the business to start recording three things.

1

Billing timeliness board
file → invoice lag

Two-thirds of the cash cycle is internal process. Cutting billing days needs no customer negotiation.

≈$3.43M
working capital

READY

2

Duplicate payment scan
same vendor + invoice no.

One-off recovery on the existing backlog, with a clear rule. It must key on the invoice number — matching on amount flags every per-container trucking charge as a false positive.

252,904
backlog exposure

READY

3

Operator performance board
volume · speed · quality

Operator attribution covers 99.7%. Skip days-to-close; use billing lag, cost-booking lag and demurrage incidence.

12 people
operators active in 2025

READY

4 Exception alert engine
rule-driven task queue

Six rules can run today: overdue billing, late cost booking, duplicate invoice numbers, balance past baseline, demurrage not passed on, negative file margin.

6 rules
all data-backed

READY

5 Collections workbench
behavioural overdue

The ledger is healthy; the prize is not chasing bad debt but pulling the $1.99M in the 1–30 day band from 15 days late to 5.

$4.69M
open receivables

READY

6 Demurrage cost attribution
customer · operator · lane · terminal

On-time pickup rate cannot be computed, but cost attribution can. Work Group B's 107 files one by one to find why they were not passed on.

945,298
cost to review

READY

7 Compliance & pickup prevention
ISF / AMS / last free day

To prevent rather than report after the fact, three inputs must be captured first: pickup and empty-return times, last free day, and ISF/AMS filing timestamps.

5,000
ISF late-filing penalty / entry

NEEDS CAPTURE

06 — NOT POSSIBLE TODAY

Four things that need a process change before any code

Listed separately so nobody builds a full reporting suite against the schema and discovers on launch day that it is empty.

On-time container pickup rate DATA MISSING

Pickup date is present on 1,187 of 134,591 containers, last free day on 12.3%, and the B/L-level `F_PickupDate` has never been filled once. This is not a query that cannot be written; the data was never captured.

```
F_PUDATE       1,187 / 134,591   0.9%
F_LastFreeDay 19,857 / 134,591  14.8%
T_OIHMAIN.F_PickupDate      0 / 85,712
```

ISF & AMS filing timeliness DATA MISSING

A late ISF costs $5,000 per entry, and that risk is currently **invisible in the system** — filing timestamps are not recorded, so there is no way to tell whether a deadline was missed.

```
F_AmsDate         0 / 17,870
F_ISFLinkID > 0   5 / 17,870
F_ClearDate       0 / 17,870
```

Payment approval trail NEVER ENABLED

The approval fields were built but never used. No payment in the system can be traced to an approver, and there is no value-based authorisation. This is an internal-control gap, not an analytics one — it deserves its own project.

```
F_Approved = '0'    73,175 / 73,175   (100%)
F_ApprovedUser           0 / 73,175
F_HoldUser               0 / 73,175
```

Salesperson assessment DATA MISSING

Salesperson attribution is filled on only 8.7% of files, so revenue, margin and retention cannot be split by salesperson. Operator assessment is unaffected.

```
F_SManID      1,549 / 17,870   8.7%
```

07 — RECOMMENDATION

Ship the billing timeliness board first

It delivers fastest, rests on the cleanest definitions, and converts directly into cash. Once it lands, management can see what this system is worth, and the remaining items will meet far less resistance. Alongside it, write the direct-versus-agency split into the data layer — every downstream metric depends on it.

Item seven (compliance and pickup prevention) should start in parallel, but as a business process change rather than analytics development. It depends on operators recording pickup times, last free days and ISF filing times. The earlier that starts, the more history there is to work with.

DATA-LAYER CONSTRAINT	REASON	SCOPE
Business model as a mandatory dimension	43% of revenue settles per cycle and cannot be attributed per file	`all per-file metrics`
Overdue measured against a behavioural baseline	57% of invoices carry zero-day terms; on-time rate is 20%	`collections & alerts`
Duty exposure managed separately from freight	duty is advanced, zero-margin and CBP-deadlined	`limits & risk`
Merge customer entities before computing exposure	the main and -DUTY accounts hide half the exposure	`all customer metrics`
Flag and isolate related parties	intercompany balances flood the collections queue	`collections workbench`

About this audit. Every figure comes from read-only queries against the TRANSKNIGHTS production database, verified 2026-09-04. Fill rates are measured over the 17,870 ocean import files opened since 2024; container-level and AP-voucher metrics note their own sample sizes.

Amounts in USD · undated figures cover 2024-01-01 onward · the working-capital figure is a linear estimate from billings, not a commitment